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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

VT2301-1- ADVANCED SEMICONDUCTOR MICROFABRICATION TECHNOLOGY
(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. In the microfabrication process, what does the term "cleanroom" refer to?
A) A room with high humidity to prevent static electricity
B) A room with controlled temperature for only cleaning processes
C) A highly controlled environment with minimal particles to prevent contamination during fabrication
D) A room for storing raw materials
2. The scaling factor of a MOS device is α . Using the constant voltage scaling model, the gate area of the device will be scaled as
A) $1/\alpha$
B) $1/\alpha^2$
C) $1/\alpha^3$
D) $1/\alpha^4$
3. In integrated circuits, the design of electronic circuits is based on the approach of use of
A) Maximum number of resistors in the circuit
B) Large sized capacitor
C) Minimum chip area irrespective of the type of components in the design
D) Use of only bipolar transistors
4. Silicon has a _____ crystal structure
A) Simple cubic
B) Face centered cubic
C) Body centered cubic
D) Diamond
5. In the Czochralski crystal growth process, the materials are heated up to
A) 950°C
B) 1000 °C
C) 1420°C
D) 1200°C
6. Which method is used for Cu deposition
A) Thermal oxidation
B) Evaporation
C) Sputtering
D) Electroplating
7. What does the "resolution" of a photoresist refer to?
A) The speed of the photolithographic process
B) The minimum feature size that can be accurately patterned
C) The thickness of the resist film
D) The color of the exposed resist
8. Undercutting is a characteristic of _____
A) Sputtering
B) Wet Etching
C) Evaporation
D) Ion milling
9. What material is commonly etched using hydrofluoric acid (HF)?
A) Silicon
B) Gold
C) Copper
D) Silicon dioxide

10. Which type of photoresist is sensitive to UV light and becomes soluble upon exposure?
 A) Positive photoresist B) Negative photoresist
 C) Electron resist D) X-ray resist
11. What does the term "etch selectivity" refer to?
 A) The speed of the etching process B) The ability to etch one material over another
 C) The uniformity of the etched surface D) The chemical composition of the etchant
12. What is the main goal of the RMG process in advanced semiconductor technology?
 A) To replace polysilicon gates with high-k dielectric materials B) To reduce the thickness of the gate dielectric
 C) To replace the gate material with metal for improved conductivity and performance D) To improve the channel mobility of the transistor
13. The dopants are introduced in the active areas of silicon by using which process?
 A) Diffusion process B) Ion Implantation process
 C) Chemical Vapour Deposition D) Either Diffusion or Ion Implantation Process
14. The general equation for Fick's Second Law is:
 A) $\partial C/\partial t = D * \partial^2 C/\partial x^2$ B) $\partial C/\partial t = - D * \partial^2 C/\partial x$
 C) $\partial C/\partial x = D * \partial^2 C/\partial t$ D) $\partial C/\partial x = - D * \partial^2 C/\partial t$
15. Which of the following processes is typically used after ion implantation to repair the damage caused to the substrate material?
 A) Annealing B) Oxidation
 C) Etching D) Lithography
16. What is the primary purpose of Chemical Mechanical Polishing (CMP) in semiconductor manufacturing?
 A) To deposit thin films on the wafer B) To etch patterns onto the wafer
 C) To smooth and flatten the wafer surface D) To clean the wafer surface
17. Which of the following is a high-k dielectric commonly used in modern semiconductor technology?
 A) Hafnium oxide (HfO_2) B) Aluminum oxide (Al_2O_3)
 C) Silicon nitride (Si_3N_4) D) Silicon dioxide (SiO_2)
18. For p-type source and drain regions, which dopant is commonly used?
 A) Boron (B) B) Arsenic (As)
 C) Phosphorus (P) D) Gallium (Ga)
19. Why is SiO_2 being replaced by materials like SiOF and SiOC in advanced semiconductor technology?
 A) They provide better mechanical properties B) They increase the metal interconnect conductivity
 C) They reduce the RC delay D) They offer better thermal conductivity
20. What is the primary purpose of the ILD in semiconductor interconnects?
 A) To improve heat dissipation B) To reduce the size of the vias
 C) To increase the interconnect density D) To provide electrical insulation between metal layers

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Analyse the effect of constant field scaling on the following parameters with the help of relevant equations.				
i.	Electric field				
ii.	specific gate capacitance				
iii.	Drain saturation current				
iv.	Power dissipation per device	8	L*2	1	4

- b) Explain the key steps involved in optical lithography. Draw the systematic process flow to realize the following structure shown in Fig.1(b) using a positive photoresist and negative photoresist.

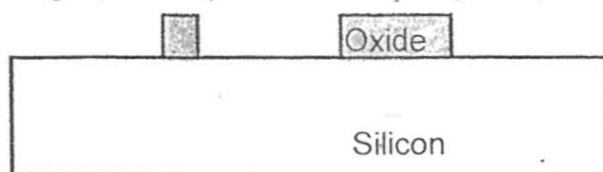


Figure. 1(b)

8 L3 2 4

2. a) Define anisotropy. Differentiate isotropic etching from anisotropic etching. Briefly explain the plasma etching process, with the help of neat diagrams and necessary equations.

8 L2 2 4

- b) With the help of neat diagrams, Explain in detail the important types of physical vapor deposition methods. Also list the advantages and disadvantages of each method.

8 L2 1 4

3. a) Explain the Electroplating process using neat diagrams. Also explain the significance of CMP process in IC fabrication with the help of neat diagrams.

8 L2 2 4

- b) Explain the process involved in diffusion used for doping of semiconductors using neat schematics? Explain the important laws governing the diffusion. Also compare diffusion process with Ion implantation process.

8 L2 2 4

Unit – II

4. a) Explain Shallow trench isolation (STI) method with neat sketches. Compare it with LOCOS process.

10 L2 4 4

- b) Compare the structure of a Planar MOSFET and a FinFET with the help of neat figures. List the advantages of FinFET.

6 L2 3 4

5. a) Explain the gate first process for gate module in detail with the help of neat diagrams. Mention its disadvantages.

8 L2 4 4

- b) Explain in detail the Self-Aligned Silicide formation with the help of neat diagrams.

8 L2 3 4

6. a) Explain the two mechanisms of diffusion in solids with the help of neat figures? Also Explain the functioning of TiN layer as a diffusion barrier layer in metal semiconductor contact technology.

8 L2 4 4

- b) Explain any four types of implants carried out during the fabrication of a transistor.

8 L2 3 4

Unit – III

7. a) Explain the trench first then via process with the help of neat diagrams and discuss its challenges.

10 L2 5 4

- b) Discuss the material challenges associated with low k dielectrics.

6 L2 5 4

8. a) What are Inter layer dielectrics (ILD)? Why are low K dielectrics needed in interconnections? Explain about the evolution of various low k dielectric materials used in fabrication.

10 L2 5 4

- b) Explain electromigration in IC fabrication and methods to reduce it.

6 L2 5 4

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
